



Workflow Management with *persistence* 1

John Reiland, CU Research Computing



Workflow Management with *persistence1*: Resources

- Website: <http://www.colorado.edu/rc>
- Documentation: <https://curc.readthedocs.io>
- Helpdesk: rc-help@colorado.edu



Slides:

https://github.com/ResearchComputing/workflow_management_persistence1

Meet the User Support Team



Layla
Freeborn



Brandon
Reyes



Andy
Monaghan



Michael
Schneider



John
Reiland



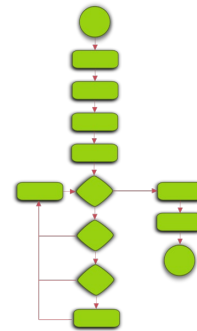
Dylan
Gottlieb



Mohal
Khandelwal

What is a workflow?

- Any/all steps needed to produce a computational result
 - Typically involves repetition, such as the same computation being performed on different pieces of data
 - Computations/analyses are done in a specific order
 - Can be sequential or parallel
- Also sometimes called a “pipeline”



What are workflow managers?

- Workflow managers are a type of software designed to set up, coordinate, launch, and monitor jobs and workflows on various types of computing environments, including HPC systems like Alpine and Blanca.
- While schedulers such as Slurm decide where and when a job runs, a workflow manager decides which jobs run, in which order, and what is needed for the whole workflow to complete successfully

Use Cases and Benefits

- Define workflow once, run many times
- Dependency management
- Parallelization
- Restart workflow at any time
- Reproducibility

When *Not* to Use Workflow Managers

- Infrequent or “one-off” workflows
- Simple workflows
- Interactive workflows
- Highly performance-sensitive tasks

Examples of Workflow Management Software

- Snakemake



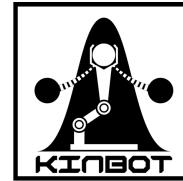
- Nextflow

- nf-core



- KinBot

- Not a traditional workflow manager, but functions similarly



Example Workflow Management Pipeline

```
#!/usr/bin/env nextflow

params.in = "$baseDir/data/sample.fa"

// Split a fasta file into multiple files
process splitSequences {

    input:
    path 'input.fa'

    output:
    path 'seq_*'

    """
    awk '/^>/{f="seq_"++d} {print > f}' < input.fa
    """
}
```

Example from <https://www.nextflow.io/example1.html>

Example Workflow Management Pipeline (continued)

```
// Reverse the sequences
process reverse {

    input:
    path x

    output:
    stdout

    """
    cat $x | rev
    """
}

// Define the workflow
workflow {
    splitSequences(params.in) \
    | reverse \
    | view
}
```

Example from <https://www.nextflow.io/example1.html>

Challenges of Workflow Managers in HPC Environments

- HPC environments are typically job-based (through schedulers such as Slurm or PBS)
 - Jobs generally come with time limits, which can be challenging to manage
 - Frequently restarting workflow manager jobs, pipelines, etc. leads to inefficiency
- Login nodes on HPC systems must not be used for any computation or long-running tasks
 - While generally lightweight, workflow managers do consume resources
 - Performance impact for others trying to access the HPC system
 - Workflow management software can run for weeks at a time

Addressing Challenges Using *persistence*¹

- A more powerful, robust, controlled-access server
 - Like a high-powered login node
 - More powerful than a standard login node, less powerful than a compute node
 - Controlled-access prevents contention with users trying to access standard resources
- No job submission required for workflow management applications
- Long-running processes allowed (as long as they don't perform computation/analyses themselves)

How to Access *persistence1*

- Submit a [Support Request Form](#) or email rc-help@colorado.edu requesting access
 - Describe your intended workflow and software usage
- Once granted, login to a terminal on the system as normal, then type `ssh persistence1`
 - You will immediately be connected to the *persistence1* server
- Set up your workflow manager and run!

More details: <https://curc.readthedocs.io/en/latest/running-jobs/persistence1.html>

Limitations of *persistence1*

- The computational tasks of your workflow *must not* be run on *persistence1*
 - *persistence1* may be used to run the workflow manager only
 - Only software relevant to workflow management and job submission is available
- SSH connections to *persistence1* can still “die” from inactivity or connection issues, which will likely terminate your processes
 - Consider software such as *screen* or *tmux* to keep sessions alive

Please take our survey!



<https://s.zoom.us/j/61PpNq5M>



Any Questions?